Trainings

Preparatory Steps

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

To reduce the possibility of serious personal injury, securely support the engine, before removing the engine support bracket.

- Drain the oil. Refer to Procedure 007-037 in Section 7 (/qs3/pubsys2/xml/en/procedures/18/18-007-037-tr.html).
- Remove the oil pan. Refer to Procedure 007-025 in Section 7 (/qs3/pubsys2/xml/en/procedures/18/18-007-025-tr.html).
- Remove the oil pan adapter cover. Refer to Procedure 007-026 in Section 7 (/qs3/pubsys2/xml/en/procedures/18/18-007-026-tr.html).
- Remove the turbocharger oil drain hoses. Refer to Procedure 010-045 in Section 10 (/qs3/pubsys2/xml/en/procedures/18/18-010-045-tr.html).
- Remove the dipstick tube and dipstick. Refer to Procedure 007-011 in Section 7

(/qs3/pubsys2/xml/en/procedures/18/18-007-011-tr.html).

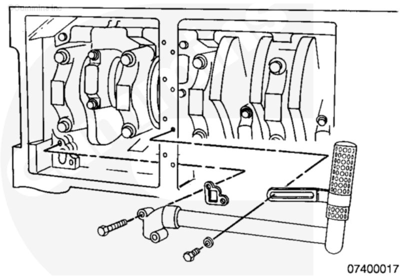
- Remove the front engine support bracket. Refer to Procedure 016-002 in Section 16 (/qs3/pubsys2/xml/en/procedures/18/18-016-002-tr.html).
- Remove the oil filters. Refer to Procedure 007-013 in Section 7 (/qs3/pubsys2/xml/en/procedures/18/18-007-013-tr.html).

Remove

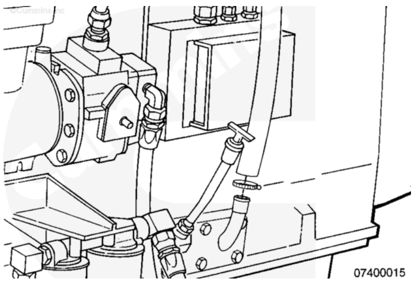
Remove the three capscrews from the suction tube.

Remove the oil pan suction tube.

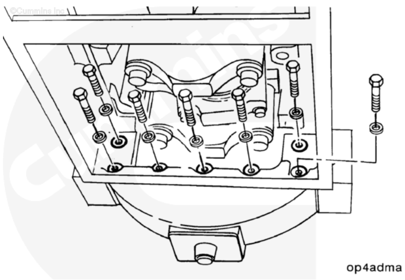
Remove and discard the gasket.



Disconnect the breather vent hose from the handhole cover.

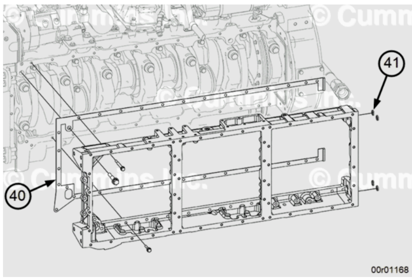


Remove the two 7/16-14 inch and five 3/8-16 inch capscrews from the flywheel housing.



⚠ WARNING ⚠

This component weighs 23 kg [50 lbs] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.

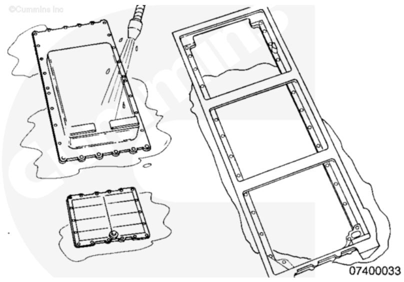


Remove all of the capscrews and the oil pan adapter.

Remove and discard the gasket (40).

Remove and discard the four bolt seals (41) from the adapter.

Clean and Inspect for Reuse



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

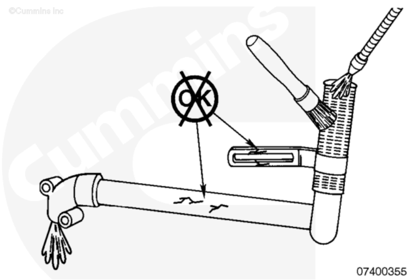
Clean the lubricating oil pan adapter with solvent, Part Number 3824421, or equivalent, that will **not** harm aluminum.

Inspect the oil pan adapter for cracks or other damage.

If cracks are suspected, use the crack detection kit, Part Number 3375432, or an equivalent kit which allows for use of the dye penetration method.

Check threaded holes for damage.

If the lubricating oil pan adapter is damaged, the lubricating oil pan adapter **must** be repaired or replaced.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

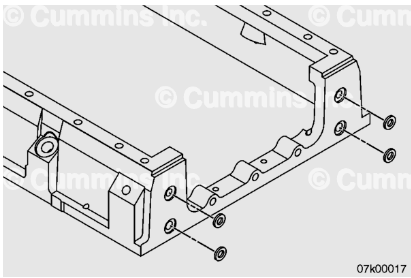
Clean the oil suction tube with solvent and dry with compressed air.

Inspect the oil suction tube for cracks or other damage.

If the oil suction tube is cracked or damaged, the oil suction tube **must** be replaced.

Install

Preferred Method

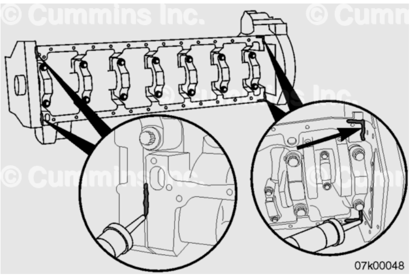


Note : The engine must be completely rolled over to perform this procedure with the Preferred Method. If a complete engine roll over is not possible, reference the Optional Method.

Note : Make sure the entire block and oil pan adapter surfaces are clean and dry of oil and debris for optimal gasket sealing and silicon adhesion.

Apply contact adhesive such as 3M™ Spray 77™ or 3M™ 4693 to hold the capscrew seals in position. Do **not** use the 3M™ Spray 77™ or 3M™ 4693 adhesive on the rectangular seal areas.

Install the four new capscrew seals on the oil pan adapter.

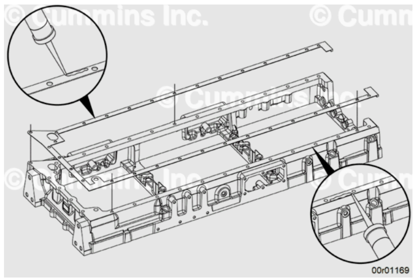


Note : Make sure the gear cover, gear cover spacer plate, and related gaskets are flush with the bottom of the block. They should not protrude more than 0.05 mm [0.002 inch] from the bottom of the block surface. If out of specification, the gear cover spacer plate alignment should be corrected to make sure the oil pan adapter gasket will conform and seal in the T-joint. Refer to Procedure 001-029 (/qs3/pubsys2/xml/en/procedures/18/18-001-029-tr.html) in Section 1 for gear cover spacer plate.

Apply Cummins® RTV, Part Number 4918882 or 3164070, or equivalent, to the bottom of the cylinder block in the following locations, as illustrated.

Apply RTV sealant, Part Number 4918882 or 3164070, or equivalent to the outside edges of the block pan rail.

Apply the new oil pan adapter gasket on the block face oil pan adapter.



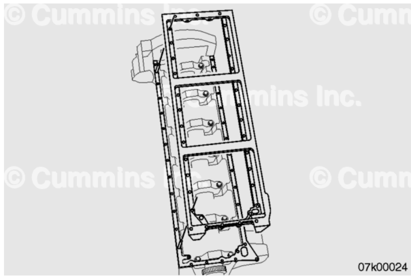
Place the new oil pan adapter gasket on the block face.

Apply RTV sealant, Part Number 4918882 or 3164070, or equivalent to outside edges of oil pan adapter gasket.

Be sure the gasket holes align with the holes in the oil pan adapter.

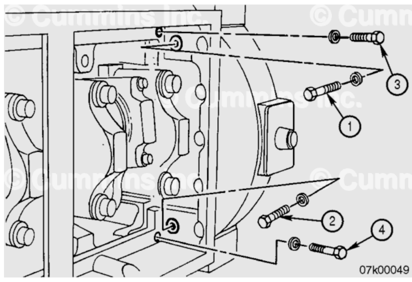
⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Position the oil pan adapter against the cylinder block and then move it toward the rear of the engine until it contacts the flywheel housing.

Lubricate the capscrew threads with clean engine oil.



Install the four washers and capscrews as shown. Push the adapter towards the block and flywheel housing while hand-tightening the capscrews.

Note : If the two block capscrews can not be started, it may be necessary to hand-tighten the two flywheel housing capscrews first so the capscrew holes in the oil pan adapter line up with the block capscrew holes.

The four capscrews shown **must** be tightened alternately and evenly using the following torque steps and sequence to be sure the adapter is seated into the corner of the block and flywheel housing simultaneously.

Torque Value:

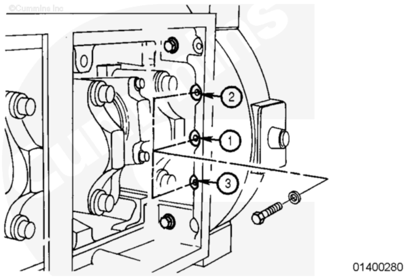
1. 15 n•m [133 in-lb]
2. 25 n•m [221 in-lb]
3. 40 n•m [30 ft-lb]
4. 48 n•m [35 ft-lb]

Lubricate the capscrew threads with clean engine oil.

Install the three capscrews located in the center of the flywheel housing and tighten them in the sequence shown. Use the steps and torque values below when tightening the capscrews.

Torque Value:

1. 25 n•m [221 in-lb]
2. 40 n•m [30 ft-lb]
3. 48 n•m [35 ft-lb]

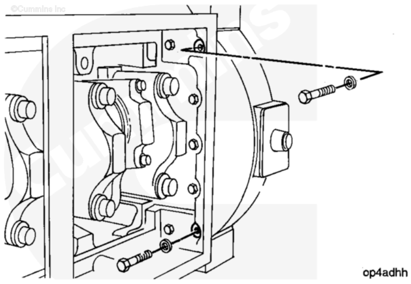


Lubricate the capscrew threads with clean engine oil.

Install the two 7/16-inch washers and capscrews shown.

Tighten the capscrews.

Torque Value: 65 n•m [48 ft-lb]



Lubricate the capscrew threads with clean engine oil.

Install all capscrews hand-tight before applying torque.

Install capscrews 24 through 28.

These capscrews thread into the front gear cover.

Install the remaining 3/8-inch washers and capscrews (1 through 23).

Install the 9/16-inch washer and capscrew (29).

Tighten all the capscrews that thread into the block.

Note : Follow the capscrew torque sequence shown. The gasket must be compressed from flywheel housing to front gear cover.

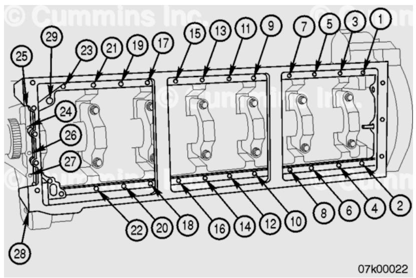
Torque Value:
Capscrews 1 - 23: 60 n•m [44 ft-lb]

Torque Value:
Capscrews 24 - 28: 48 n•m [35 ft-lb]

Torque Value:
Capscrew 29: 150 n•m [111 ft-lb]

Repeat the final torque value and sequence on all oil pan adapter capscrews.

Use the previous four stepblocks as a guide.



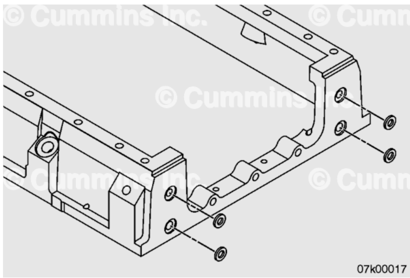
Optional Method

Note : Make sure the entire block and oil pan adapter surfaces are clean and dry of oil and debris for optimal gasket sealing and RTV adhesion.

Apply contact adhesive such as 3M™ Spray 77™ or 3M™ 4693 to hold the capscrew seals in position.

Do **not** use the 3M™ Spray 77™ or 3M™ 4693 adhesive on the rectangular seal areas.

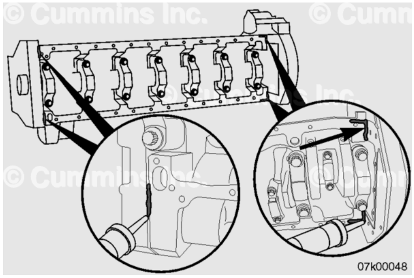
Install the four new capscrew seals on the oil pan adapter.



Note : Make sure the gear cover, gear cover spacer plate, and related gaskets are flush with the bottom of the block. They should not protrude more than 0.05 mm [0.002 inch] from the bottom of the block surface. If out of specification, the gear cover spacer plate alignment should be corrected to make sure the oil pan adapter gasket will conform and seal in the T-joint. See Procedure 001-029 in Section 1 for gear cover spacer plate.

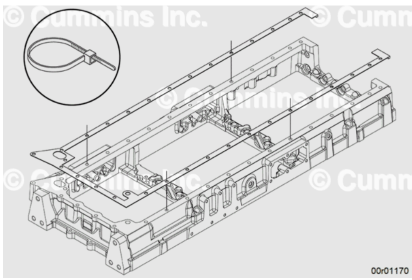
Apply Cummins® RTV, Part Number 4918882 or 3164070, or equivalent, to the bottom of the cylinder block in the following locations as shown:

- the joint where the cylinder block meets the front gear housing
- the joint where the cylinder block meet the flywheel housing.
    - Apply in an L pattern as shown.



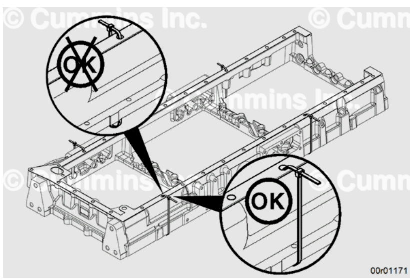
Install the new oil pan adapter gasket onto the oil pan adapter.

To ease in assembly, plastic wire ties should be looped though a capscrew hole, toward the outside of the oil pan adapter, to hold the gasket in place.

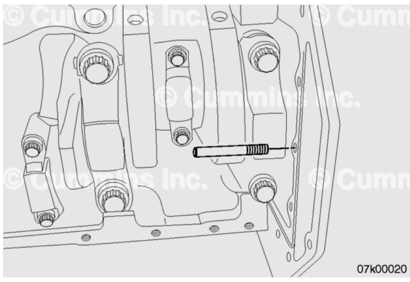


Install wire ties in a minimum of four places on the outside of the oil pan adapter as shown.

Note : Wire ties improperly looped toward the inside of the oil pan adapter can damage the seal bead when the ties are removed and result in a leak.

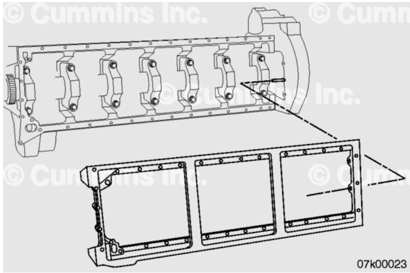


Install a 3/8" guide stud into the flywheel housing bolt hole to assist in the installation of the oil pan adapter.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Install the oil pan adapter.

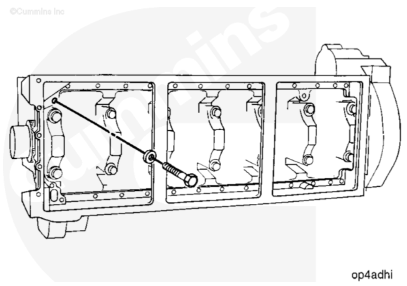
Align the oil pan adapter with the guide stud.

Position the oil pan adapter against the cylinder block face and the flywheel housing.

Be sure to check the rectangular seal after the adapter has been set into place, and again when the adapter is tight. Seals may slip out of place. Seals **must** be checked prior to the RTV curing.

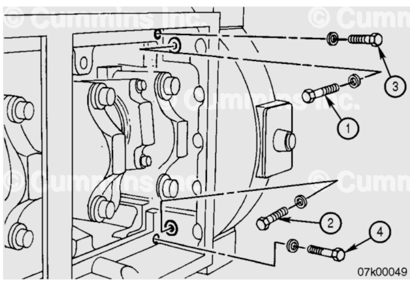
Lubricate the oil pan adapter 9/16-inch capscrew with clean engine oil.

Install and finger-tighten the 9/16-inch washer and capscrew.



Lubricate the capscrew threads with clean engine oil.

Install the four washers and capscrews as shown. Push the adapter toward the block and flywheel housing while hand-tightening the capscrews.



Note : If the two block capscrews can not be started, it may be necessary to hand-tighten the two flywheel housing capscrews first so the capscrew holes in the oil pan adapter line up with the block capscrew holes.

The four capscrews shown **must** be tightened alternately and evenly to be sure the adapter is pulled evenly using the torque steps and sequence below to make sure the adapter is seated into the corner of the block rails and flywheel housing simultaneously.

Remove the wire ties.

Use the following steps while tightening the oil pan adapter capscrews.

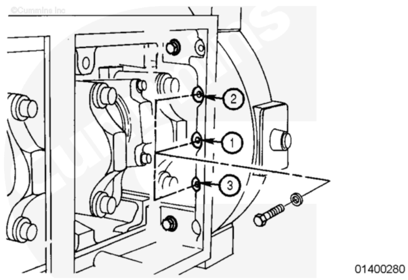
Torque Value:

1. 15 n•m [133 in-lb]
2. 25 n•m [221 in-lb]
3. 40 n•m [30 ft-lb]
4. 48 n•m [53 ft-lb]

Remove the guide stud from the flywheel housing.

Lubricate the capscrew threads with clean engine oil.

Install the three capscrews located in the center of the flywheel housing and tighten them in the sequence illustrated. Use the steps and torque values below when tightening the capscrews.



Torque Value:

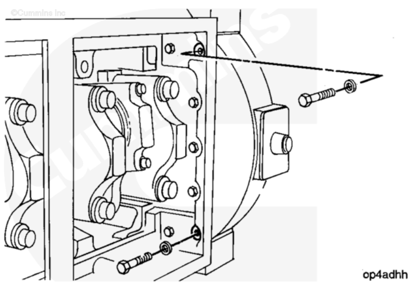
1. 25 n•m [221 ft-lb]
2. 40 n•m [30 ft-lb]
3. 48 n•m [35 ft-lb]

Lubricate the capscrew threads with clean engine oil.

Install the two 7/16-inch washers and capscrews shown.

Tighten the capscrews.

Torque Value: 65 n•m [48 ft-lb]

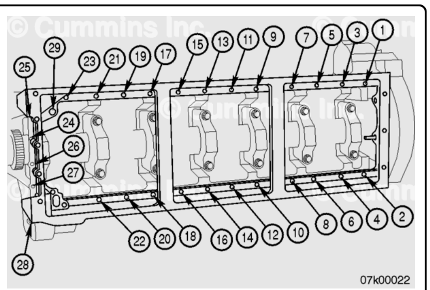


Lubricate the capscrew threads with clean engine oil.

Install all capscrews hand-tight before applying torque.

Install capscrews 24 through 28. These capscrews thread into the front gear cover.

Install the remaining 3/8-inch washers and capscrews (1 through 23).



Tighten all the capscrews that thread into the block.

Note : Follow the capscrew torque sequence illustrated. The gasket must be compressed from flywheel housing to front gear cover.

Torque Value:
Capscrews 1 - 23: 60 n•m [44 ft-lb]

Torque Value:
Capscrews 24 - 28: 48 n•m [35 ft-lb]

Torque Value:
Capscrew 29: 150 n•m [111 ft-lb]

Repeat the final torque value and sequence on all oil pan adapter capscrews.

Use the previous four step blocks as a guide.

Finishing Steps

- Install the oil filters. Refer to Procedure 007-013 in Section 7 (/qs3/pubsys2/xml/en/procedures/18/18-007-013-tr.html).
- Install the front engine support bracket. Refer to Procedure 016-002 in Section 16 (/qs3/pubsys2/xml/en/procedures/18/18-016-002-tr.html).
- Install the dipstick tube and dipstick. Refer to Procedure 007-011 in Section 7 (/qs3/pubsys2/xml/en/procedures/18/18-007-011-tr.html).
- Install the turbocharger oil drain hoses. Refer to Procedure 010-045 in Section 10 (/qs3/pubsys2/xml/en/procedures/18/18-010-045-tr.html).
- Install the oil pan adapter cover. Refer to Procedure 007-026 in Section 7 (/qs3/pubsys2/xml/en/procedures/18/18-007-026-tr.html).
- Install the oil pan. Refer to Procedure 007-025 in Section 7 (/qs3/pubsys2/xml/en/procedures/18/18-007-025-tr.html).
- Fill the engine with oil. Refer to Procedure 007-037 in Section 7 (/qs3/pubsys2/xml/en/procedures/18/18-007-037-tr.html).

- Operate the engine to normal operating temperature and check for leaks.